

NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

SC4001 Neural Networks and Deep Learning

Group Project – Group 14

Do ECG Heartbeat Classifiers Generalise to Unseen Patients?

A Controlled Study of a 1D CNN on MIT-BIH

Name	Matriculation Number	Role
Hou Junning	U2421246K	Data and features
Nguyen Khanh	U2520033E	Models and training
Huang Yitian	U2423017H	Evaluation and report

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1 Introduction

With around 100,000 beats in a single day of recording, long-term monitoring with the electrocardiogram (ECG) relies on automatic heartbeat classification. For this task, the MIT-BIH Arrhythmia Database [1], [2] is the standard benchmark, and its beats are grouped into the AAMI classes normal (N), supraventricular ectopic (S), ventricular ectopic (V), fusion (F) and unknown (Q) [3].

However, reported results depend heavily on how the beats are split. If beats from all recordings are pooled and divided at random (the *intra-patient* paradigm), beats from the same person appear in both the training and the test set, so a model can succeed by recognising the person rather than the arrhythmia. Under this paradigm, deep models report near-perfect scores, for instance 100% S sensitivity at 96.5% positive predictivity [4]. Once the training and test recordings are disjoint (the *inter-patient* split of de Chazal et al. [5]), the task becomes much harder. Their classifier found 75.9% of S beats at a positive predictivity of only 38.5%, and Sellami and Hwang report 99.48% accuracy intra-patient but 88.34% inter-patient [6]. Beyond ECG, the same pattern appears across clinical machine learning. For instance, record-wise validation often overestimates accuracy compared with subject-wise validation [7], and leakage between training and test data has produced over-optimistic conclusions in many fields [8]. Since a deployed classifier always meets a new patient, only a patient-wise evaluation answers the clinically relevant question.

Nonetheless, the patient-wise benchmark has a difficulty of its own, as abnormal beats are rare and concentrated in a few people. In the standard test set (DS2, with 49,661 beats after our preprocessing), S beats make up only 3.7% of all beats, and a single recording, 232, contains 75% of them. Recording 213 likewise contains 93% of the test F beats. As a result, predicting N for every beat already scores 89.0% accuracy, even though its macro-F1 is only 0.24.

Hence, this project asks **how much a 1D convolutional neural network (CNN) loses when it is evaluated on patients it has never seen, and whether inexpensive changes recover the rare abnormal classes.** We train one baseline on both a random beat split and a subject-disjoint patient split. We then test four changes, each aimed at a suspected cause of the gap, namely class-weighted and focal losses for class imbalance [9], [10], waveform augmentation for differences in recording conditions [11], [12], and RR-interval features for beat timing. Timing matters because S beats are mostly early relative to the local rhythm, while their shape resembles that of N beats. Our contributions are:

- A controlled measurement of the gap, in which macro-F1 falls from 0.874 to 0.429 and S and F F1 fall from 0.74 and 0.81 to zero. Unlike the conventional split, which ignores that records 201 and 202 come from the same person, our patient split is fully subject-disjoint.
- A comparison of four fixes over three seeds, with per-class sensitivity, positive predictivity and F1, per-patient results and patient-bootstrap intervals. None of the fixes recovers S detection. Class weighting appears to recover it, but only by labelling whole patients as S.
- A failure analysis showing that RR features teach the network to rank S beats correctly *within* a patient (AUROC 0.994 in record 232), even though it never predicts S, and that a decision threshold set on other patients does not fix this.
- A data analysis explaining why timing does not transfer, as 62% of training S beats lie inside tachycardia runs, where they are not early, whereas most test S beats are isolated early beats.

2 Review of Existing Techniques

2.1 Heartbeat classification on MIT-BIH

The MIT-BIH Arrhythmia Database contains 48 half-hour, two-channel ECG recordings from 47 subjects, sampled at 360 Hz [13]. De Chazal et al. [5] established a widely used inter-patient evaluation protocol, dividing 44 non-paced recordings into DS1 and DS2, each containing 22 recordings. DS1 is used for training and model selection, while DS2 is reserved for independent testing. Their classifier combined ECG morphology and heartbeat-interval features. Subsequent feature-based approaches improved generalisation through feature selection [14] and combinations of temporal and morphological descriptors [15].

Class imbalance. ECG datasets are highly imbalanced, with normal beats dominating minority classes, making accuracy an unreliable performance metric. Buda et al. [16] demonstrated the negative effects of class imbalance on CNNs. Class-weighted losses penalise errors on rare classes more heavily; Cui et al. [9] proposed weighting based on the effective number of samples. Focal loss [10] reduces the contribution of easily classified examples. ECG-specific approaches include batch-weighted loss [6] and class-balancing augmentation [17]. Our study compares inverse-frequency weighting (E2) and focal loss (E3) under patient-independent evaluation.

2.2 Timing features and data augmentation

Heartbeat classification can combine waveform morphology with temporal context. de Chazal et al. [5] used morphology and heartbeat-interval features with separate development and evaluation record sets. This motivates comparing an isolated-waveform CNN with one that also consumes RR features. Our four-class neural model and four-feature timing vector differ from their classifier, so we do not claim to reproduce its reported performance.

Augmentation offers a complementary way to test robustness to recording variation. Nonaka and Seita [18] studied augmentation for single-lead atrial-fibrillation classification, a different task from our heartbeat classification. Raghu et al. [19] found that augmentation effects depend on the ECG prediction task and proposed the learned TaskAug policy. We instead fix gain, noise, baseline and alignment perturbations, keeping architecture and evaluation patients unchanged. These studies motivate testing augmentation; they do not establish that our transforms improve unseen-patient performance. We test these questions in E4 and E5.

2.3 Evaluation protocols and patient-level uncertainty

An evaluation is only meaningful if the relationship between training and test data matches the relationship between training data and future use [7]. However, even the conventional inter-patient split is not fully subject-disjoint, as records 201 (DS1) and 202 (DS2) come from the same person. PhysioNet also advises keeping them in one partition for subject-independent work [13]. Apart from the split, how results are summarised also matters. While ANSI/AAMI EC57 emphasises record-by-record reporting [3], headline inter-patient results are single figures per class, pooled over DS2. Such pooled figures are dominated by a few recordings. Beats from one patient are not independent either, so intervals that treat beats as independent samples are too narrow. For clustered data like this, the standard bootstrap resamples whole clusters, here patients [20], [21], [22]. For these reasons, alongside the pooled metrics, we report per-record results, results without the dominant records, and patient-bootstrap intervals.

3 Data and Methods

3.1 Data and preprocessing

We used the MIT-BIH Arrhythmia Database v1.0.0 [13], excluding paced records 102, 104, 107 and 217 and retaining 44 recordings. Original waveform, header and expert annotation files were verified against the publisher's SHA-256 checksums. Beat symbols were grouped into four superclasses: N={N,L,R,e,j}, S={A,a,J,S}, V={V,E} and F={F}, encoded as 0–3. N therefore includes non-ectopic types beyond normal sinus beats. Q-class targets and non-beat annotations were excluded.

We selected MLII by channel name, including the second channel in record 114. Each reference annotation supplied a 256-sample window spanning [−128, +128) samples at the original 360 Hz, approximately 0.711 s. We applied no resampling or bandpass filter. Excluding 15 Q targets, 40 boundary-crossing target windows and 48 further targets without previous/next RR context left 100,630 labelled beats; 2,991 non-target annotations were ignored. All experiments use this retained pool. For each protocol, a scalar waveform mean and population SD were fitted on training beats only and frozen for validation/test. RR features were standardised column-wise in the same way. Annotated centres make this a beat-classification experiment conditional on known beat locations, rather than a complete detection system.

3.2 Partitions

E0 uses a stratified 80/20 development/test split, with 20% of development reserved for validation: approximately 64/16/20 overall. The data seed is fixed at 42 across all model seeds. Subjects and neighbouring-beat context may overlap across these random partitions.

E1–E5 use `patient_strict`, based on the conventional DS1/DS2 record lists [5]. DS1 records 108, 118, 203 and 223 form validation; DS2 is reserved for final evaluation. Since records 201 and 202 belong to the same person [13], we remove record 201 from training. The resulting 17 training, four validation and 22 test subjects are disjoint. This is an explicit modification of the conventional protocol, which remains available as a reference. Table 1 gives the actual retained counts. All model seeds share these partitions and their training-fitted normalisation; protocol prefixes are never mixed. Changing split also changes cohort composition and training size, so the E0/E1 gap cannot isolate patient overlap as its sole cause.

3.3 RR features and augmentation

E5 adds four RR features: previous RR, next RR, the mean of up to ten preceding RR intervals including the current previous interval, and the previous RR divided by that mean, termed the *prematurity index*. The first three are measured in seconds; the index is dimensionless, with values below one indicating an early beat

Table 1: Retained beats by partition and class. Random is used by E0; strict patient is used by E1–E5.

Protocol	Partition	Records	Beats	N	S	V	F
Random	Train	44	64,404	57,627	1,778	4,485	514
Random	Validation	44	16,101	14,407	445	1,121	128
Random	Test	44	20,125	18,008	556	1,401	160
Strict patient	Train	17	39,394	35,719	640	2,640	395
Strict patient	Validation	4	9,614	8,472	175	950	17
Strict patient	Test	22	49,661	44,218	1,836	3,219	388

relative to the local rhythm. Features are computed from consecutive beat annotations within each recording, retaining Q beats as timing context but excluding rhythm markers. Intervals never cross record boundaries and class identities are not feature inputs. All four columns, including the index, are standardised using training statistics. Next-RR requires future context, so E5 is offline classification.

For E4, each training waveform independently receives gain scaling in [0.9, 1.1], Gaussian noise with SD 0.02, sinusoidal baseline wander with amplitude 0.05 and frequency in [0.1, 0.5] Hz with random phase, and an integer shift in [−8, 8] samples, each with probability 0.5. Shifts use edge padding. Additive magnitudes are in training-standardised units. Transform randomness is determined by model seed, epoch and row index. Labels and RR features stay fixed because these perturbations model measurement variation. Validation/test waveforms are never augmented; E0–E4 do not consume RR features.

3.4 Model and training

We implemented a lightweight 1D convolutional neural network (CNN) with 13,508 trainable parameters. The model uses a three-block convolutional encoder, followed by global average pooling and a linear classification head producing four logits (N, S, V, F). For E5, four standardised RR-interval features are concatenated with the CNN representation before classification, increasing the parameter count to 13,524.

Models were trained using AdamW with a learning rate of 10^{-3}, weight decay of 10^{-4}, and batch size 256 on an NVIDIA RTX 5050 GPU. Training was limited to 30 epochs, with early stopping after four consecutive epochs without improvement in validation macro-F1. The best-performing checkpoint was retained. Each experiment used three random seeds (42, 43, 44). E2 employed inverse-frequency class-weighted cross-entropy with weights 0.025 (N), 1.389 (S), 0.337 (V), and 2.250 (F). E3 used focal loss with $\gamma = 2$.

The validation set contained only 17 F beats and 175 S beats, of which 169 originated from records 118 and 223. This limited minority-class representation makes validation macro-F1 sensitive to individual predictions and record-specific characteristics. Early stopping may therefore have selected checkpoints before the models adequately learned minority-class patterns, potentially affecting generalisation.

3.5 Evaluation

As Fig. 1 shows, abnormal beats are concentrated in a few records. In the training set, record 209 holds 60% of the S beats and record 208 holds 94% of the F beats, while in the test set, record 232 holds 75% of the S beats and record 213 holds 93% of the F beats. Hence, pooled test scores mostly describe a few patients, and we designed our evaluation around this.

We report per-class sensitivity (Se, recall), positive predictivity (+P, precision) and F1, together with the macro-F1 over the four classes, as mean ± SD over the three seeds. Test predictions were exported only once, from the frozen checkpoints. Before any test beat was read, the processed arrays passed the data audit, and every checkpoint reproduced its stored validation macro-F1 on our copy of the data (largest difference 3×10^{-4}, from GPU/CPU rounding). Since two records dominate the test S and F beats, we also report S without record 232 and F without record 213 as sensitivity analyses. Finally, to show how much a result depends on which patients are tested, we resampled the 22 test records with replacement 2,000 times, using the same draws for every model. For each metric and for its paired difference from E1, we report the 95% percentile interval.

4 Experiments and Results

4.1 Experiments and hypotheses

E0 (random split) and E1 (patient split) use the baseline CNN with cross-entropy, while E2–E5 each change one element relative to E1. The changed elements are class-weighted loss (E2), focal loss (E3), augmentation (E4) and RR features (E5). Before any training run, we wrote down five hypotheses:

- the gap is large and concentrated in S and F (H1)
- weighted and focal losses raise rare-class sensitivity at the cost of +P (H2)

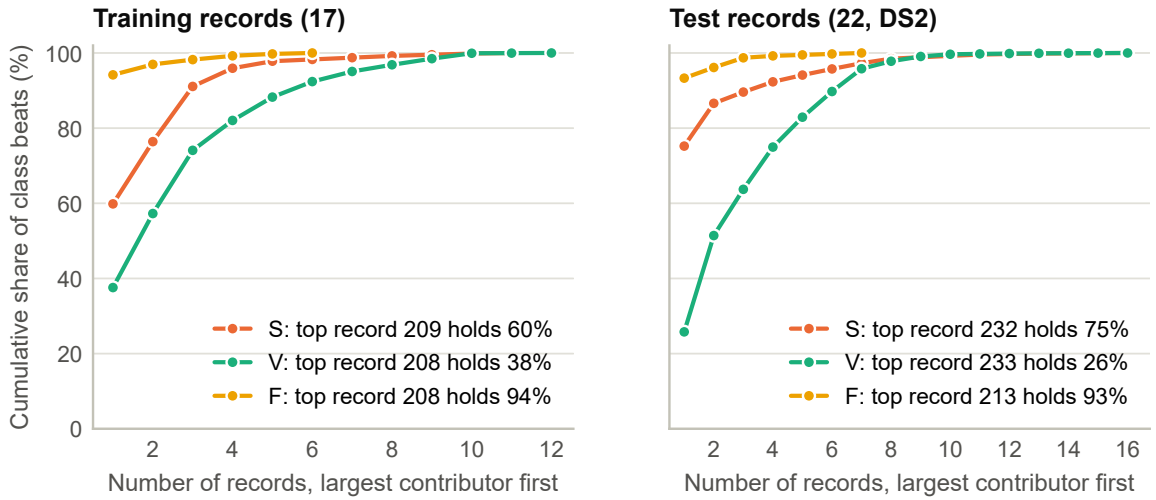


Figure 1: Concentration of abnormal beats: cumulative share of each class's beats, records sorted by contribution. Left: the 17 training records; right: the 22 test records.

- augmentation helps little (H3)
- RR features give the largest S gain (H4)
- no conclusion about F will hold (H5).

A sixth hypothesis, on autoencoder pretraining (E6), was not tested. Although the encoders were pretrained, no supervised E6 classifier was trained (Appendix B). Two analyses in Section 4.4 were added after the first test evaluation, so they are marked *post-hoc*, and no model or threshold was selected on test data.

4.2 The generalisation gap (H1: supported)

Table 2: Main results on the full test set (mean ± SD over three seeds). n/a: the class was never predicted. *Only one seed predicted any S beat.

Exp	Model (test set)	Macro-F1	S Se	S +P	S F1	V F1	F F1
–	always predict N (patient)	0.236	0	n/a	0	0	0
E0	baseline (random split)	0.874 ± 0.009	0.618 ± 0.027	0.910 ± 0.017	0.736 ± 0.023	0.955 ± 0.005	0.814 ± 0.010
E1	baseline (patient)	0.429 ± 0.002	0.000 ± 0.000	n/a	0.000 ± 0.000	0.756 ± 0.008	0.000 ± 0.000
E2	+ weighted loss	0.371 ± 0.067	0.830 ± 0.060	0.115 ± 0.013	0.201 ± 0.018	0.629 ± 0.102	0.004 ± 0.002
E3	+ focal loss	0.444 ± 0.017	0.000 ± 0.000	n/a	0.000 ± 0.000	0.811 ± 0.067	0.001 ± 0.002
E4	+ augmentation	0.433 ± 0.017	0.000 ± 0.000	n/a	0.000 ± 0.000	0.771 ± 0.061	0.001 ± 0.003
E5	+ RR features	0.445 ± 0.030	0.001 ± 0.001	0.167*	0.001 ± 0.002	0.799 ± 0.092	0.027 ± 0.047

The same network that reaches macro-F1 0.874 on a random beat split reaches only 0.429 on unseen patients (Table 2, Fig. 2). However, accuracy hides this. It falls only from 0.980 to 0.923, which is still 3.3 points above the 0.890 of always predicting N. As H1 predicted, the loss is concentrated in the rare classes. S F1 falls from 0.736 to 0 and F F1 from 0.814 to 0, whereas V F1 falls less, from 0.955 to 0.756. On unseen patients, the baseline never predicts S. Of the 1,836 test S beats, 98.6% are called N, and the F beats of record 213 are split evenly between N and V.

4.3 Effect of the fixes (H2–H5)

Table 3: Change from E1 with 95% patient-bootstrap intervals (2,000 resamples of the 22 test records, seed-averaged), and S and F F1 without their dominant records.

Exp	Δ Macro-F1	Δ S F1	Δ V F1	Δ F F1	S F1 without 232	F F1 without 213
E2	−0.058 [−0.135, +0.042]	+0.201 [+0.008, +0.473]	−0.127 [−0.227, +0.016]	+0.004 [0.000, +0.017]	0.027 ± 0.009	0.001 ± 0.001
E3	+0.015 [−0.001, +0.034]	0	+0.056 [−0.006, +0.126]	+0.001 [0.000, +0.020]	0	0.002 ± 0.003
E4	+0.004 [−0.003, +0.016]	0	+0.015 [−0.014, +0.049]	+0.001 [0.000, +0.051]	0	0.008 ± 0.013
E5	+0.016 [+0.004, +0.053]	+0.001 [0.000, +0.012]	+0.043 [+0.012, +0.079]	+0.027 [0.000, +0.172]	0.006 ± 0.010	0.001 ± 0.001

- **H2, partly supported.** Class weighting (E2) raises S sensitivity from 0 to 0.83. However, S +P is only 0.115,

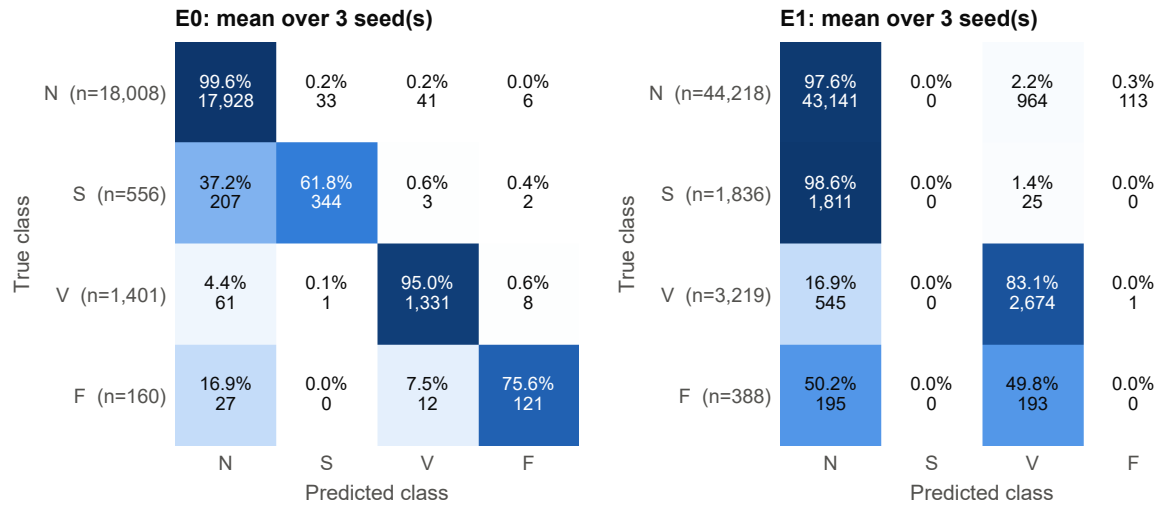


Figure 2: Confusion matrices of the same CNN on a random beat split (E0, left) and on unseen patients (E1, right), mean over three seeds. Cells show the percentage of the true class and the number of beats.

so about eight of every nine S predictions are false. Macro-F1 falls by 0.058 (the interval includes zero), and V F1 and accuracy (0.530 ± 0.155) fall with it. Without record 232, the S gain almost disappears (S F1 0.027), and Section 4.4 shows that even the gain in record 232 is not beat detection. As for focal loss (E3), it does not change S or F, and its small gains are in V.

- **H3, supported.** Augmentation (E4) changes macro-F1 by only +0.004 [−0.003, +0.016] and detects no S beats.
- **H4, not supported.** E5 predicts S for almost no test beat (S Se 0.001). Although its macro-F1 gain, +0.016 [+0.004, +0.053], is the only macro-F1 interval that excludes zero, the gain comes from V (+0.043) rather than S.
- **H5, supported.** F F1 is at most 0.03 for every patient-split model. Even the largest, E5's 0.027, has an interval from 0 to 0.172 and comes from a single seed. Training F beats also come almost entirely from record 208, and test F beats from record 213.

Moreover, seed spreads are large for the patient-split models (for instance ± 0.09 in V F1 for E5), and the bootstrap intervals are wide. That is to say, the results depend heavily on which patients are tested. Fig. 3 shows S sensitivity for each test record.

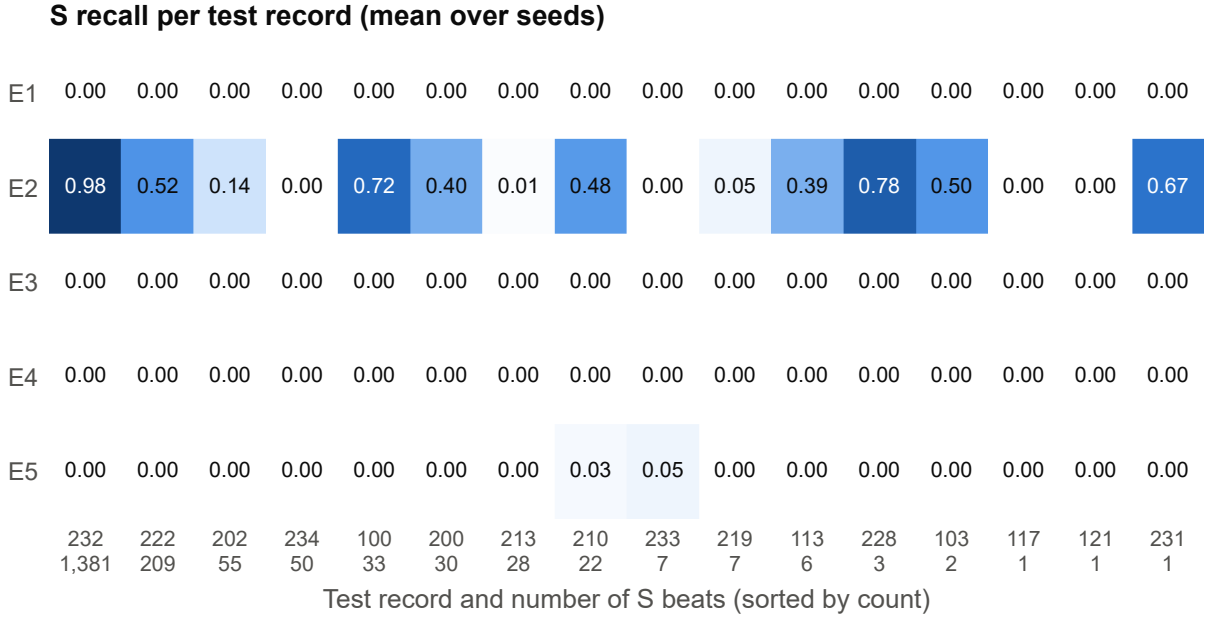


Figure 3: S sensitivity for each test record (mean over seeds), records sorted by their number of S beats (second line of the labels). E2's sensitivity in record 232 comes with S predictions for almost every beat of that record (Section 4.4).

4.4 Failure analysis

Error locations. For E1, two records account for 86% of the S errors, namely record 232, where 1,379 S beats (99.9% of its S beats) are called N, and record 222, where all 209 S beats are called N. Apart from these, the

large error cells are record 213 (363 N beats called V, with its F beats split between N and V) and record 233 (202 V beats called N, 24%).

Learning S on the training patients. Using training and validation data only, we found that the selected checkpoints of E1, E3, E4 and two of the three E5 seeds detect at most 3% of the S beats in their own training set. Early stopping kept epochs 1–5 in 14 of the 15 patient-split runs (epoch 9 in the other), compared to 18–29 for E0. This happened because validation macro-F1 stopped improving almost at once, while the validation loss of E1 rose from the first epochs (Fig. 4). Nonetheless, the models that did fit S still fail on new patients. E2 detects 85–90% of its training S beats and E5 (seed 42) detects 61%, yet they find at most 8% of the S beats of the validation patients. Hence, there are two separate failures. Model selection stops before S is learned, and S learned on the training patients does not transfer.

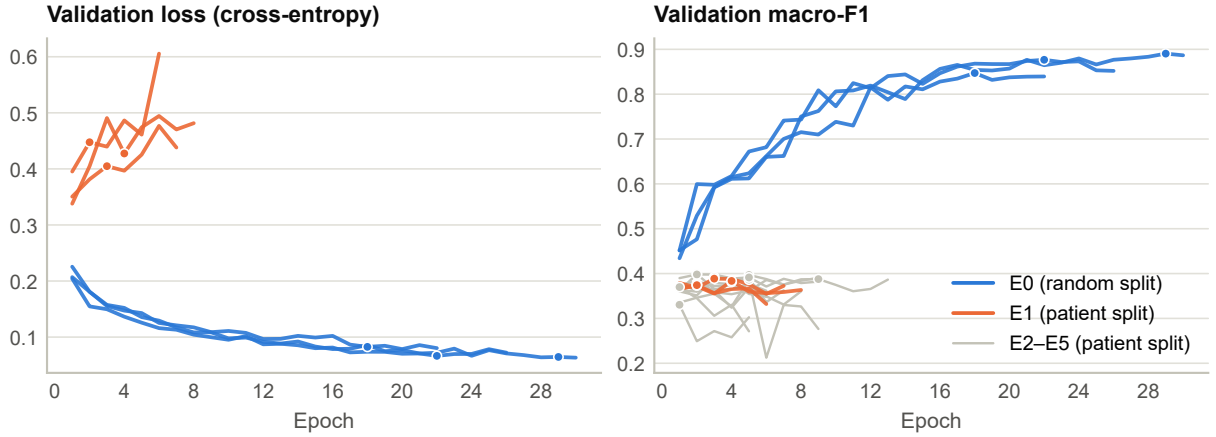


Figure 4: Validation loss (E0 and E1, cross-entropy) and validation macro-F1 (all experiments) per epoch; dots mark the epoch kept by early stopping. On the patient split, validation loss rises after the first epochs and macro-F1 barely improves after the first few.

Patient-level labelling by E2 *(post-hoc)*. Class weighting makes E2 call 98% of all beats in record 232 S, including 99% of that record's 397 N beats. Since 78% of the record's beats are S, this produces most of the S "gain". The same behaviour also produces the false positives. In record 212, 92% of the N beats (2,530 of 2,746) are called S, as are 80% of those in record 222 and 68% of those in record 100. Within record 232, E2's S probability separates S from N beats no better than chance (AUROC 0.48 ± 0.06).

Within-patient ranking of S beats *(post-hoc)*. Table 4 measures how well each model's S probability ranks S beats above all other beats, both across all test beats pooled and within each record. Pooled AUROC looks moderate for every patient-split model (0.75–0.86), but within records, E1–E4 are at or below chance (median 0.46–0.48), so their pooled score separates patients, not beats. E5 is the exception. Within record 232, its S probability separates S from N almost perfectly (AUROC 0.994), and the median over records is 0.895, yet the argmax never selects S.

Table 4: *(Post-hoc.)* AUROC of the S probability for S versus all other beats, mean over seeds. Within records: median over the 8 test records with at least 10 S and 10 other beats.

Exp	Pooled	Within records (median)	Within record 232	Share of record 232 predicted S
E0 (random split)	0.963	0.943	0.948	0.775
E1	0.753	0.467	0.429	0.000
E2	0.859	0.478	0.482	0.983
E3	0.863	0.475	0.461	0.000
E4	0.762	0.457	0.419	0.000
E5	0.863	0.895	0.994	0.000

Threshold transfer *(post-hoc; thresholds chosen on validation patients only)*. An obvious fix is a better decision threshold, so we predicted S whenever the S probability exceeded the threshold that maximised S F1 on the four validation records. However, this threshold does not transfer to the test patients for any model. For E1, E2 and E4, the threshold labels most test beats S (S Se 0.94–1.00, +P 0.04–0.07). E3 reaches S F1 0.21 ± 0.11, but with within-record ranking at chance, this again labels patients rather than beats. For E5, the outcome depends on the seed (S Se 0.62 ± 0.45, +P 0.11 ± 0.05, S F1 0.15 ± 0.08), and macro-F1 is no better than with the argmax (0.420 against 0.445). This suggests that the scores are informative within a patient but not comparable between patients.

Timing of S beats in training and test. In the training set, 62% of S beats (395 of 640) occur inside supraventricular tachycardia runs, mostly in record 209, where they are not early relative to the preceding beats. Only

48.9% of training S beats have a prematurity index below 0.9, compared to 87.7% of test S beats (Fig. 5). Within record 232, S beats correlate with that record's N beats at r = 0.975 in shape, so their timing is what differs.

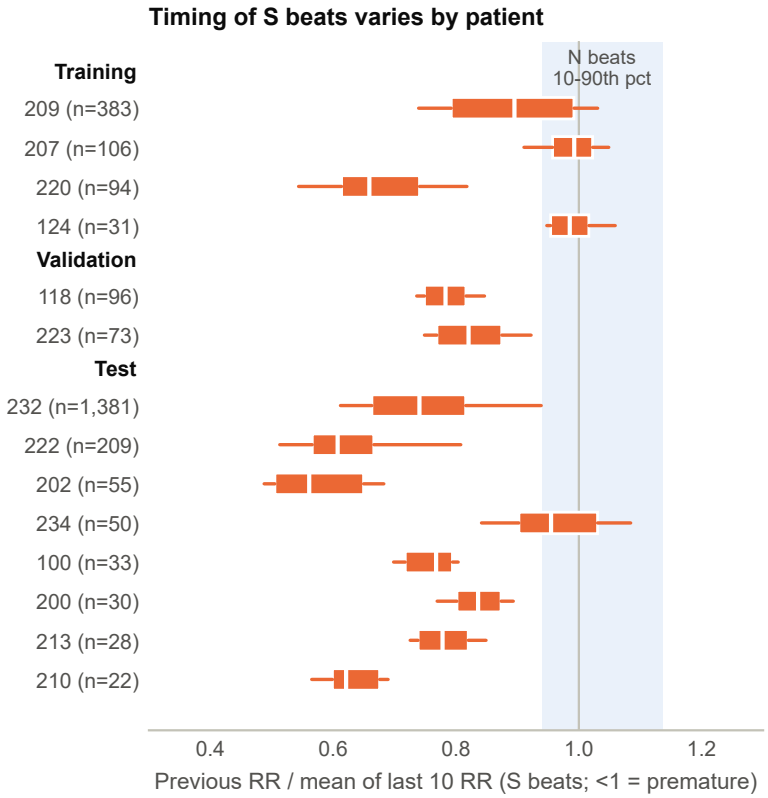


Figure 5: Prematurity index of S beats per record (previous RR divided by the mean of the last ten RR intervals; below 1 means early): median, interquartile range and 10th–90th percentiles. Training records 209 and 207 have S beats that are not early; most test S beats, led by record 232, are early.

5 Discussion and Limitations

Dependence on the split. The same 13.5k-parameter CNN, trained in the same way, scores macro-F1 0.874 when beats are split at random but only 0.429 when the test patients are new, and on new patients it detects no S or F beats. While this matches the direction reported in the literature [4], [6], it also shows how far the random-split number overstates what a deployed model would do.

Failure of the inexpensive fixes. Reweighting the loss (E2) did not teach the network what an S beat looks like in a new patient. Instead, it taught the network to call whole patients S. Although the classes are highly imbalanced, imbalance is not what blocks generalisation. As expected, augmentation (E4) did not help either, since it simulates gain, noise and baseline changes, while the S beats of record 232 differ from their N beats in timing rather than shape. RR features (E5) did teach the right cue, since within record 232 the network ranks S above N almost perfectly. Nonetheless, no test beat crosses the decision boundary, and a threshold chosen on the validation patients does not help, because S scores are not comparable between patients. As Section 4.4 shows, the training data teach a weaker timing cue than the test patients need.

Pooled versus within-patient metrics. For E1–E4, the gap between pooled and within-patient AUROC (0.75–0.86 against 0.46–0.48) shows that a pooled ranking metric can look reasonable even when a model only separates patients with many S beats from patients with few. Since within-patient analysis exposes this, we recommend reporting it alongside pooled scores.

Model selection under patient shift. Early stopping on validation macro-F1 chose epochs 1–5 in 14 of 15 patient-split runs, before the network had learned S even on its training patients. This early stop is itself a result of the patient shift. That is, learning S on the training patients did not raise validation macro-F1, because the validation patients' S beats differ. Where S was learned (E2 and one E5 seed), validation S sensitivity stayed at or below 0.08, so the failure to generalise does not depend on the early stop. Despite this, E3–E5 partly compare under-trained models, and a selection rule that does not rely on four validation patients would make the comparison cleaner and fairer.

Limitations. Although every comparison in this report is patient-wise, it must be noted that our evidence has five limits. (i) *Few patients.* There are 22 test patients, and two of them hold most of the S beats (75% in record 232) and F beats (93% in record 213). Hence, S and F conclusions describe a handful of individuals, and a

22-record bootstrap is itself approximate [22]. (ii) *Weak validation set.* Four patients with 175 S and 17 F beats guide early stopping and threshold choice. (iii) *Statistics.* We used three seeds per configuration, with SDs up to 0.155 (E2 accuracy), intervals instead of significance tests, and no correction for comparing four fixes. (iv) *Data.* We used one lead and annotated beat locations instead of a detector, the next-RR feature is offline, paced records and the Q class are excluded, and there is no external test database. (v) *Post-hoc analyses.* The ranking and threshold analyses were designed after we saw the first test results. They explain the failure and were not used to select any model.

Future work. Rather than more features, the within-patient result suggests that the missing step is per-patient calibration. This could mean normalising RR features by each patient's own rhythm, adapting the threshold with a few labelled beats from the new patient, as in patient-specific training [23], or unsupervised test-time adaptation. Each of these needs a validation design with more patients, for instance leave-one-patient-out on DS1.

6 Conclusion

On MIT-BIH, a 1D CNN that scores macro-F1 0.874 on a random beat split scores only 0.429 on patients it has not seen and detects none of their S or F beats. Four inexpensive fixes did not change this. We conclude that class weighting raised S sensitivity to 0.83 only by labelling entire patients S (S +P 0.115), that focal loss and augmentation had no effect on S, and that RR features gave the only reliable macro-F1 gain (+0.016, through V). Additionally, the main effect of RR features was hidden – the model learned to rank S beats correctly within each patient (AUROC 0.994 in record 232) but never predicted them, and a threshold chosen on other patients did not fix this. Hence, patient-wise evaluation should report per-patient results and within-patient ranking alongside pooled scores, because pooled scores can hide both failures and partial successes. The next step is to calibrate decisions per patient.

AI Use Statement

We used generative AI tools in all three roles, and each member checked the output of the tool they used.
Hou Junning used OpenAI Codex for the data pipeline, autoencoder pretraining, documentation and report drafting. Outputs were verified using source-data checksums, protocol tests, encoder audits and checks against primary literature.
Nguyen Khanh used OpenAI ChatGPT (GPT-5) for environment set up, model creation, finding documentation, code evaluation, Readme and report drafting. Code was checked by running the training script and the unit tests; text was checked against the cited papers.
Huang Yitian used Claude Code (Claude Opus 5.5 and Sonnet 5.5) for evaluation code, figures, reference checking and report drafting. Code was checked with unit tests, reported numbers against the results tables, and references against their sources.
Our team chose the research question, the subject-disjoint protocol and the experiments, decided how to report the early-stopping problem, and interpreted the results. Every result comes from experiments we ran. All code is our own, built on PyTorch, NumPy, pandas, matplotlib and the WFDB Python package. No third-party model code or externally pretrained models were used, and the E6 encoders were pretrained by us.

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A Reproducing the results

Data preparation. From a fresh clone, run the following from the repository root; CPU is sufficient for preparation and audit:

```
python -m pip install -r requirements.txt
python -m src.prepare
python -m src.data audit
python -m unittest discover -s tests -v
```

The repository includes the 44 original raw recordings and reference metadata; large NPZ arrays are rebuilt locally. `src.prepare` verifies raw checksums, builds missing arrays with matching local metadata, and audits complete caches. Use `patient_strict_{train,val,test}.npz` for E1–E5 and `random_{train,val,test}.npz` for E0. Install a suitable PyTorch build to execute the model/E6 tests; they are skipped in a data-only environment.

NPZ byte hashes may differ across environments. Preserve any mismatched existing cache and rebuild into a new empty directory, for example `python -m src.prepare --output-dir data/processed_v2`. Pass the new directory to prediction with `--data-dir` and to evaluation with `--processed-dir`; do not commit locally regenerated reference metadata or edit hashes to bypass an audit failure.

Model training. All six experiments (E0–E5) were trained using AdamW with a learning rate of 10^{-3}, weight decay of 10^{-4}, and batch size of 256. Training was limited to 30 epochs, with early stopping after four consecutive epochs without improvement in validation macro-F1. Each experiment was repeated with three seeds (42, 43, 44). E0 used the random split, while E1–E5 used the `patient_strict` protocol. The reported runs used an NVIDIA GPU; training also runs on CPU with `--device cpu`, but more slowly. To reproduce the experiments, run from the repository root:

```
python -m src.train --experiments all --seeds 42 43 44 \
    --patient-protocol patient_strict --epochs 30 --patience 4 \
    --batch-size 256 --lr 0.001 --weight-decay 0.0001 --device auto
```

The best checkpoint for each run is saved under `checkpoints/E*/seed*.pt`, with training histories and configurations stored in `results/E*/`.

The evaluation is reproduced from the repository root with:

- `python -m src.predict` exports test predictions from the committed checkpoints. CPU is enough, but PyTorch is needed (`pip install -r requirements-pretrain.txt`).
- `python -m src.evaluate describe-data` writes the dataset facts behind Table 1, Fig. 1 and Fig. 5 to `results/data_analysis/`.
- `python -m src.evaluate summarise --reference E1` writes Tables 2–4, Figs. 2–4 and Fig. C1 to `results/summary/`.

B Autoencoder pretraining (E6)

A convolutional autoencoder was pretrained on training waveforms only, separately for the conventional and strict patient protocols with seeds 42, 43 and 44. Three stride-two convolutional stages with 16, 32 and 64 channels, GroupNorm and ReLU map each 256-sample beat to a 64 × 32 representation; a mirrored transposed-convolution decoder reconstructs the waveform. Training minimised MSE using AdamW (learning rate 0.001, weight decay 0.0001, batch size 256) for ten fixed epochs without augmentation. Labels, validation/test waveforms and validation-based epoch selection were excluded. The final-epoch encoders, complete histories and checksums are versioned in `artifacts/pretraining/`; all six exports were independently audited.

To audit the exports or repeat strict-protocol pretraining in a new output directory:

```
python -m pip install -r requirements-pretrain.txt
python -m src.pretrain_audit
python -m src.pretrain --protocol patient_strict --seeds 42 43 44 \
    --epochs 10 --output-dir local_pretraining
```

The continuation character above is for Bash/Colab; on other shells, place the training command on one line. A controlled E1/E6 comparison requires identical encoder/head architectures and matching training-data provenance, with only encoder initialisation changed. The exact archive-hash transfer check can reject arrays rebuilt in another environment. **No supervised E6 classifier was trained; H6 remains not tested.** Lower reconstruction MSE establishes optimisation of the pretraining objective, not improved classification or unseen-patient generalisation.

C Bootstrap differences

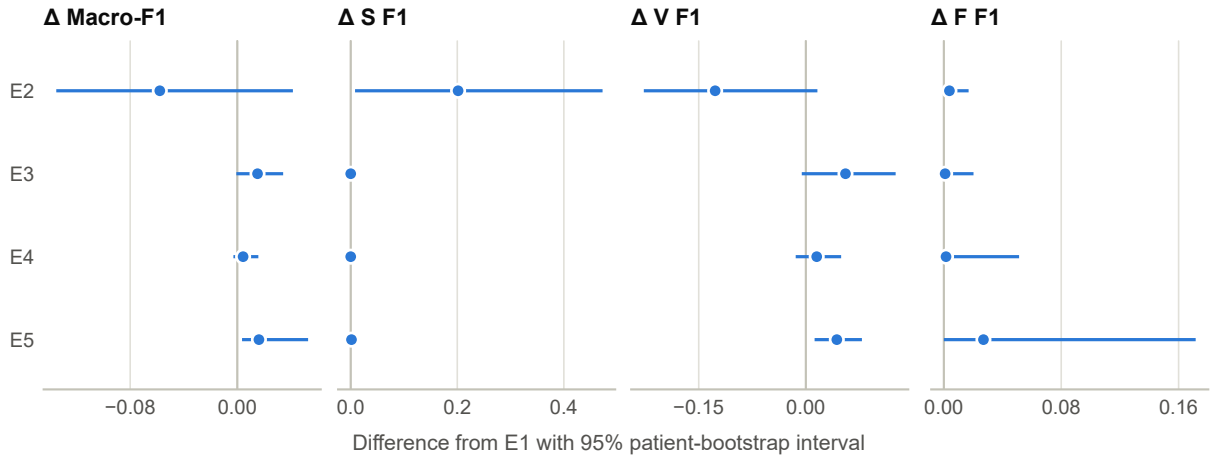


Figure C1: Change from E1 with 95% patient-bootstrap intervals (the data of Table 3).